TDUK 18th April 2018

Planning Underground in Singapore

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AECOM

What is Singapore like to live there?

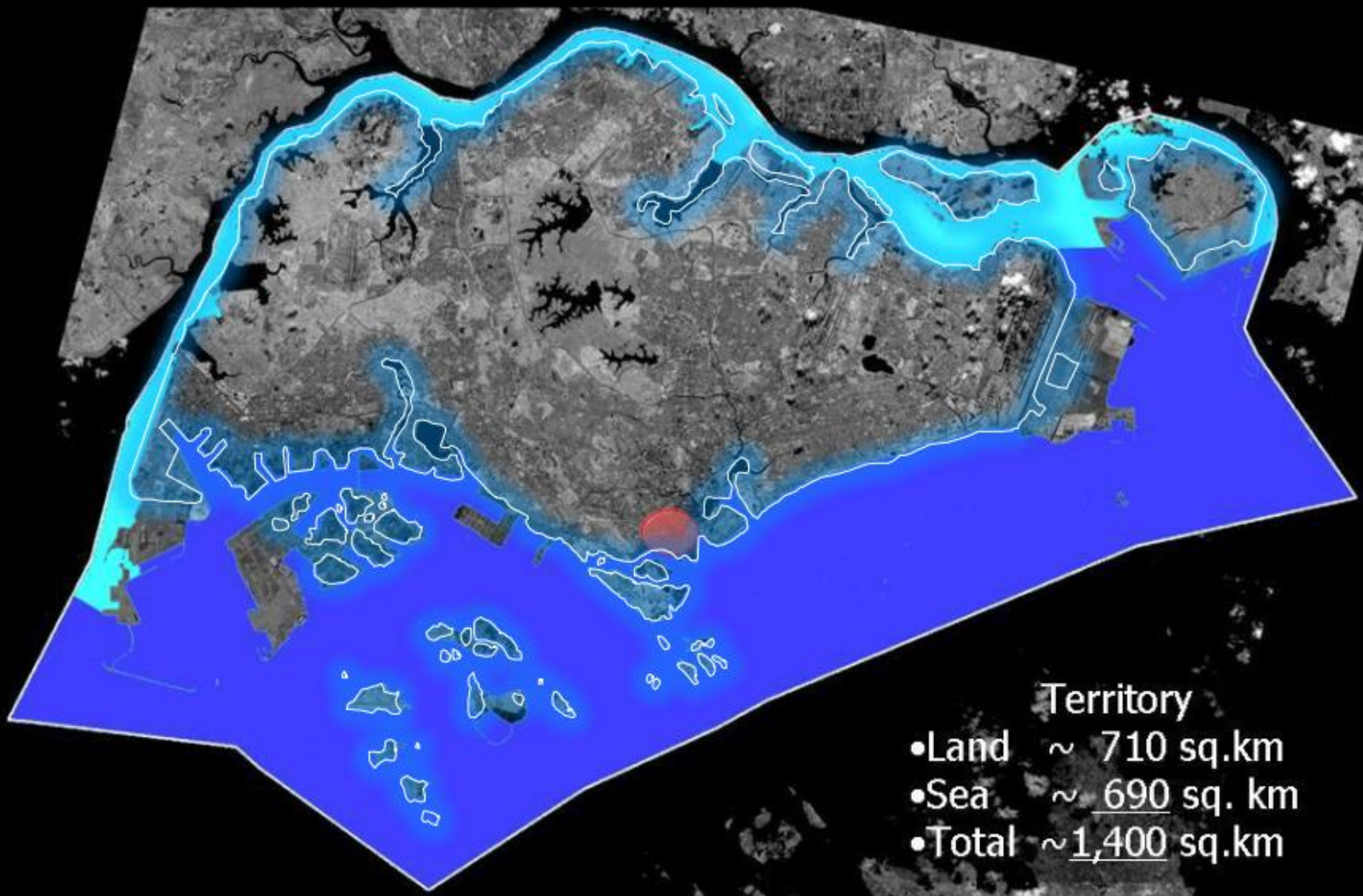
Singapore has evolved as an attractive city in which to live, work, play, educate, medical care:-

- Good housing,
- Attractive infrastructure, facilities, parkland,
- Efficient commercial, industrial areas,
- Conserved rain forest and wetlands, etc.

High rise buildings and open parkland



Territory of Singapore ~ 1,400 sq.km



Territory
- Land ~ 710 sq.km
- Sea ~ 690 sq. km
- Total ~1,400 sq.km

Half open space : Half intensively developed



Reality: 80% live in Public Housing



Strategic Planning

Singapore has evolved as an attractive city in which to live:-

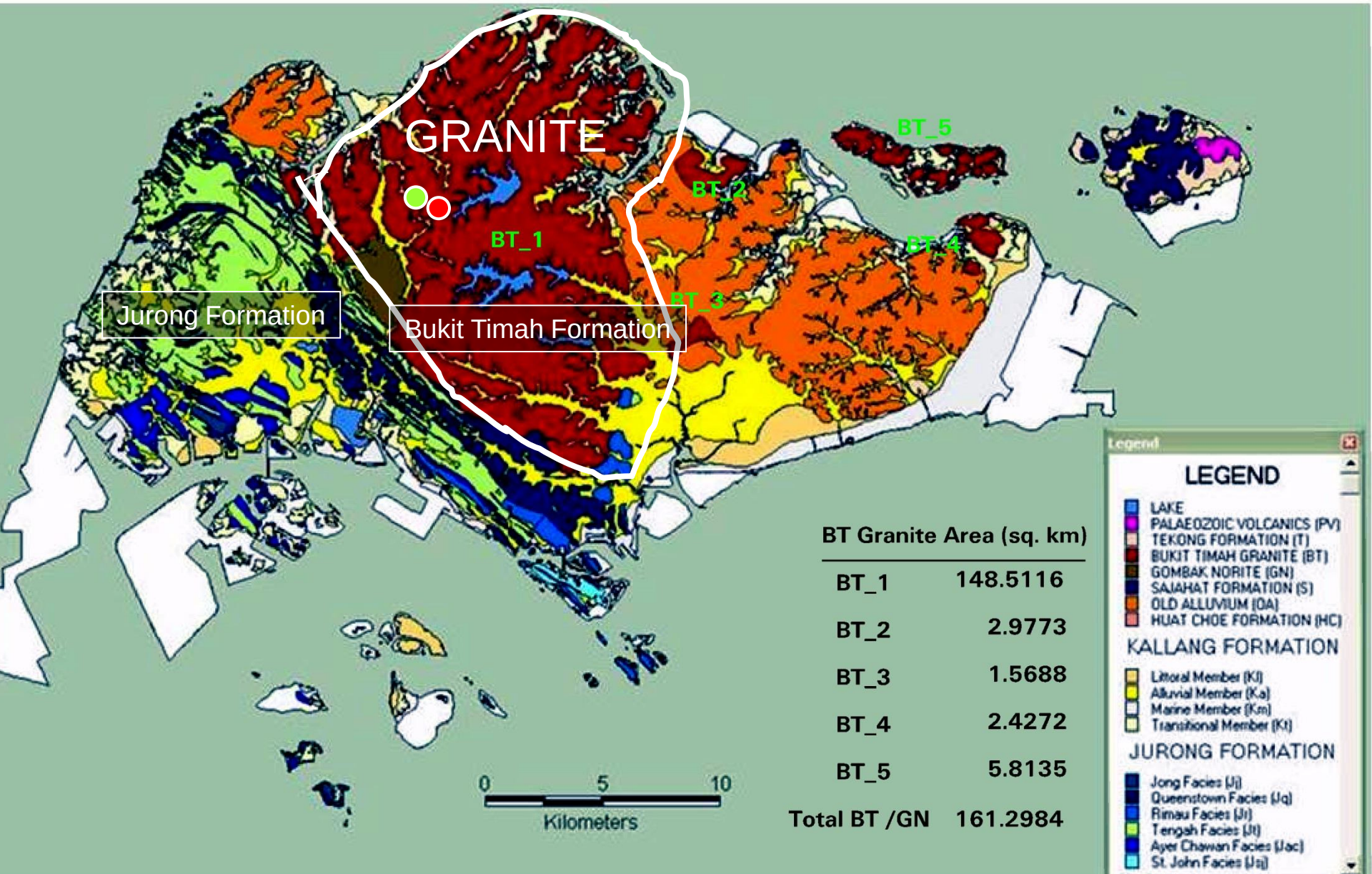
- Good housing,
- Attractive infrastructure, facilities, parkland,
- Efficient commercial, industrial areas,
- Conserved rain forest and wetlands, etc.

If the surface environment is to be sustained or improved then needed developments

Require creation of new land.

- Reclamation : the price of sand
- Underground : the quality of rock,

Geological Map of Singapore

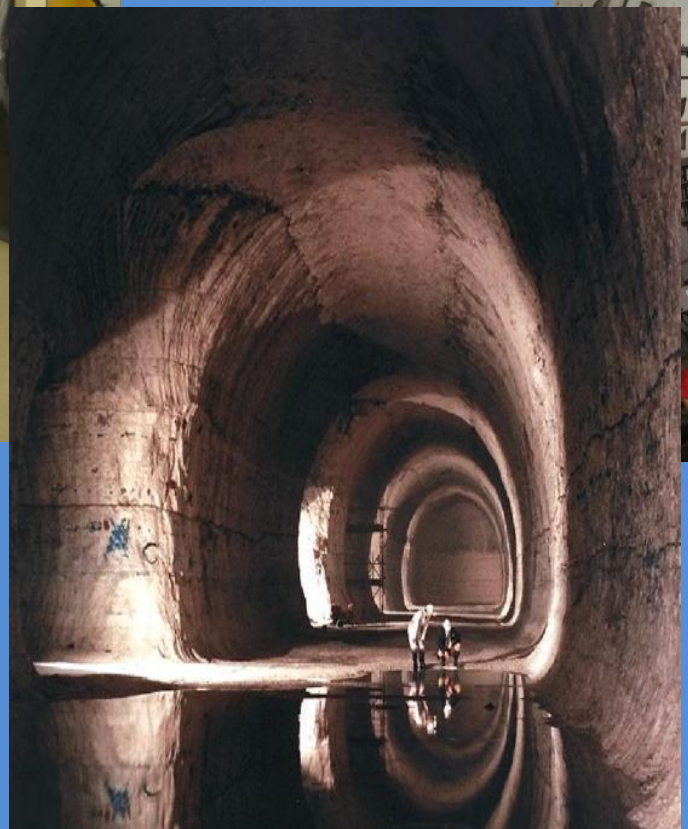


BT Granite Area (sq. km)	
BT_1	148.5116
BT_2	2.9773
BT_3	1.5688
BT_4	2.4272
BT_5	5.8135
Total BT /GN	161.2984

Current Underground Uses

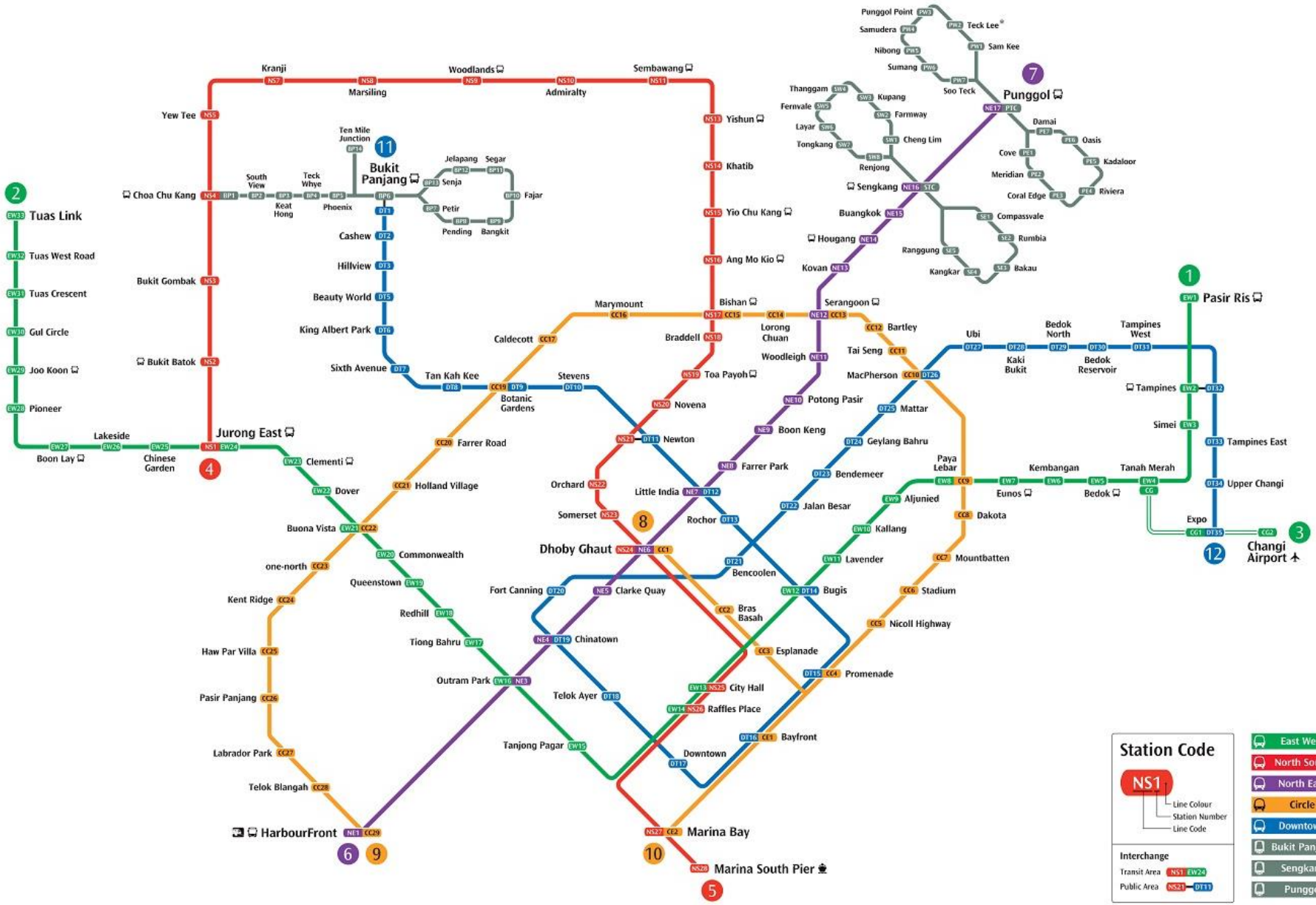
basements tunnels and caverns







Railway System Map - Singapore



MRT : Underground travel in Singapore



Underground railway connects to basements and shopping areas



MBFC Link Mall leading to Downtown MRT Station DTL

Hong Kong … links below ground



Connection to basement arcade



Connection between stations

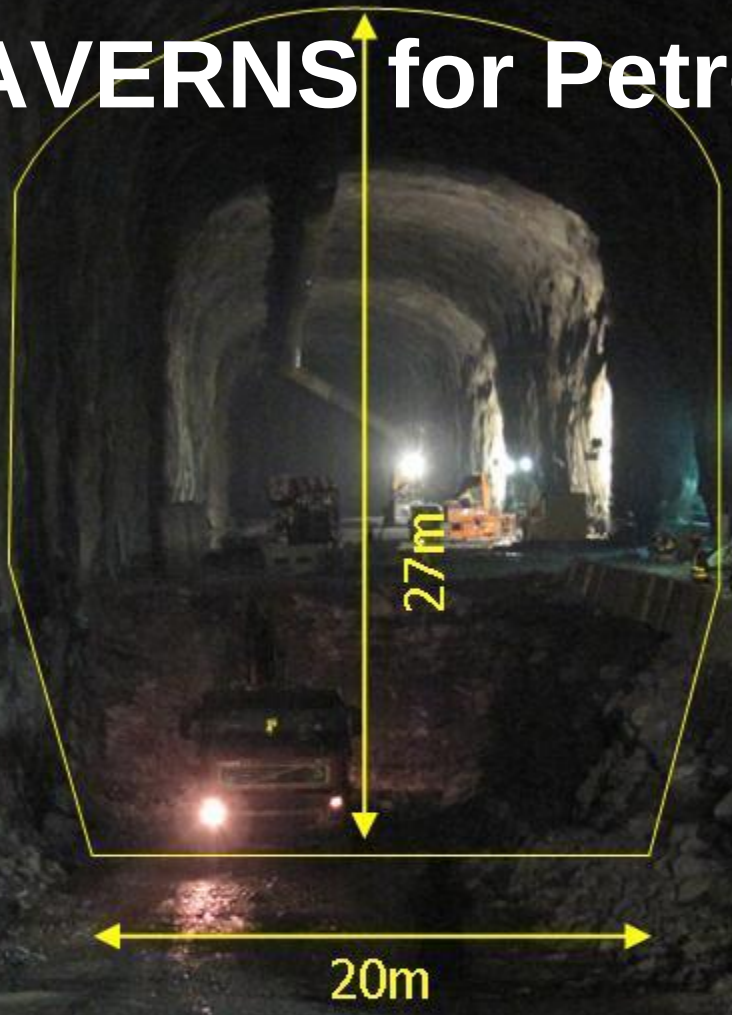


Regional interconnections

Get pedestrians off the streets



ROCK CAVERNS for Petroleum



- 9 storeys high
- Length = 340 m
- Exacavation using Drill and Blast Method
- Volume per storage cavern
  = 165,000m3
  (~ 64 Olympic-sized swimming pools)

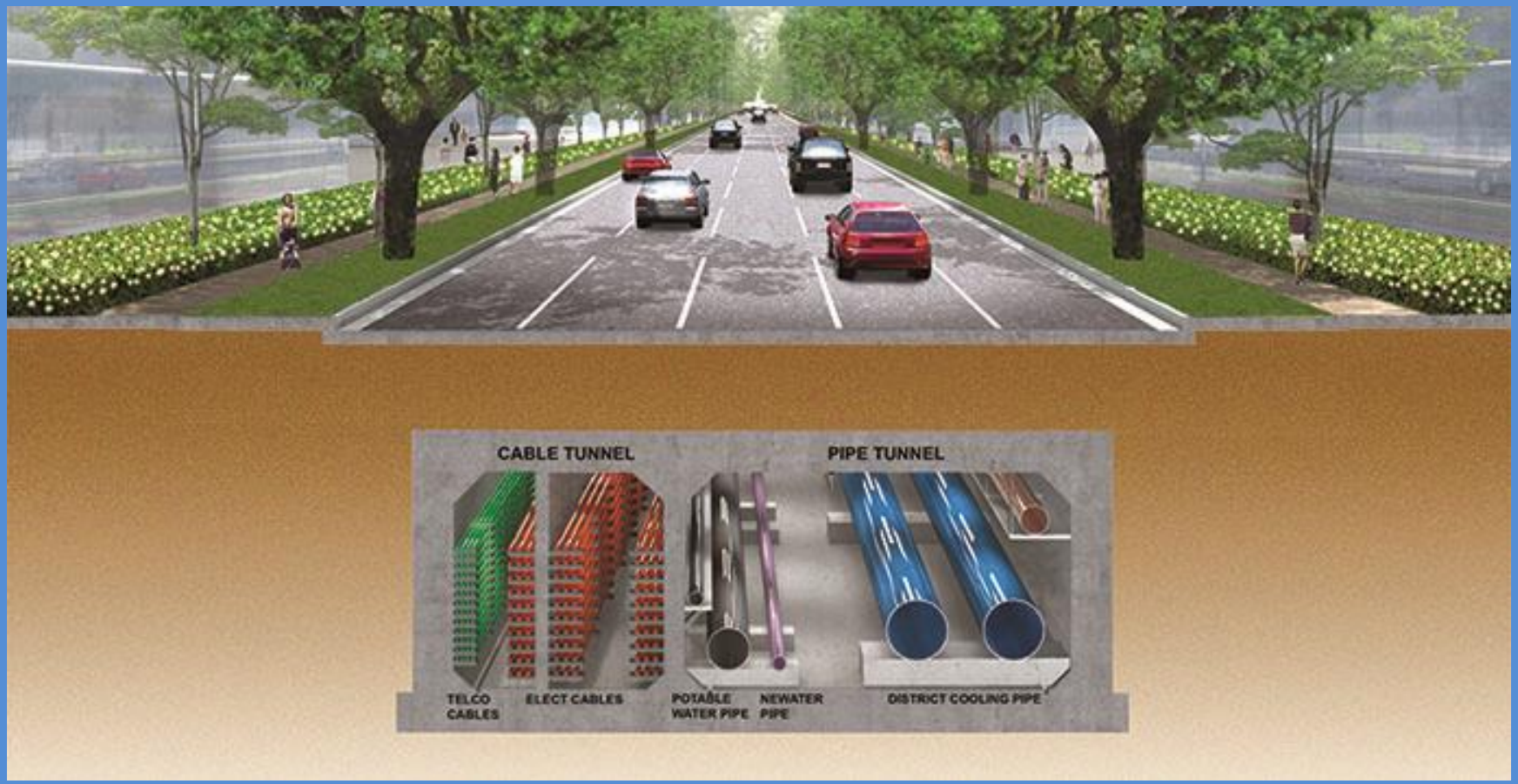
150 Metres Below the Seabed

Singapore:
Temporary shaft access to oil storage caverns



Photo courtesy of JTC

Singapore:
Combined utility tunnels, conceptual



Combined utility cable tunnel in Singapore



Facilitation/implementation

Law:

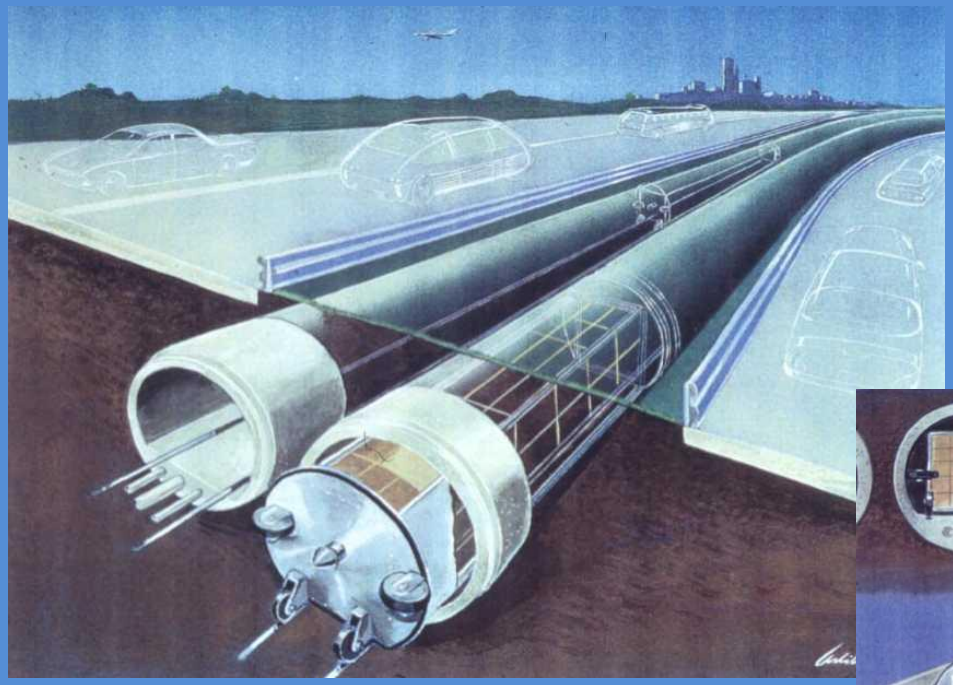
Resume land below 30m

Regulations:

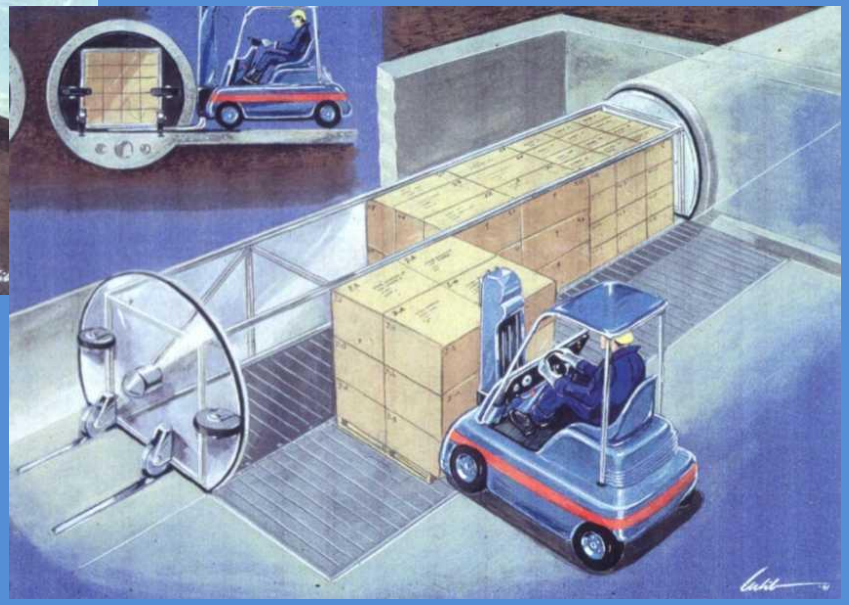
Connections between adjacent lots

Opportunities for co-development below ground

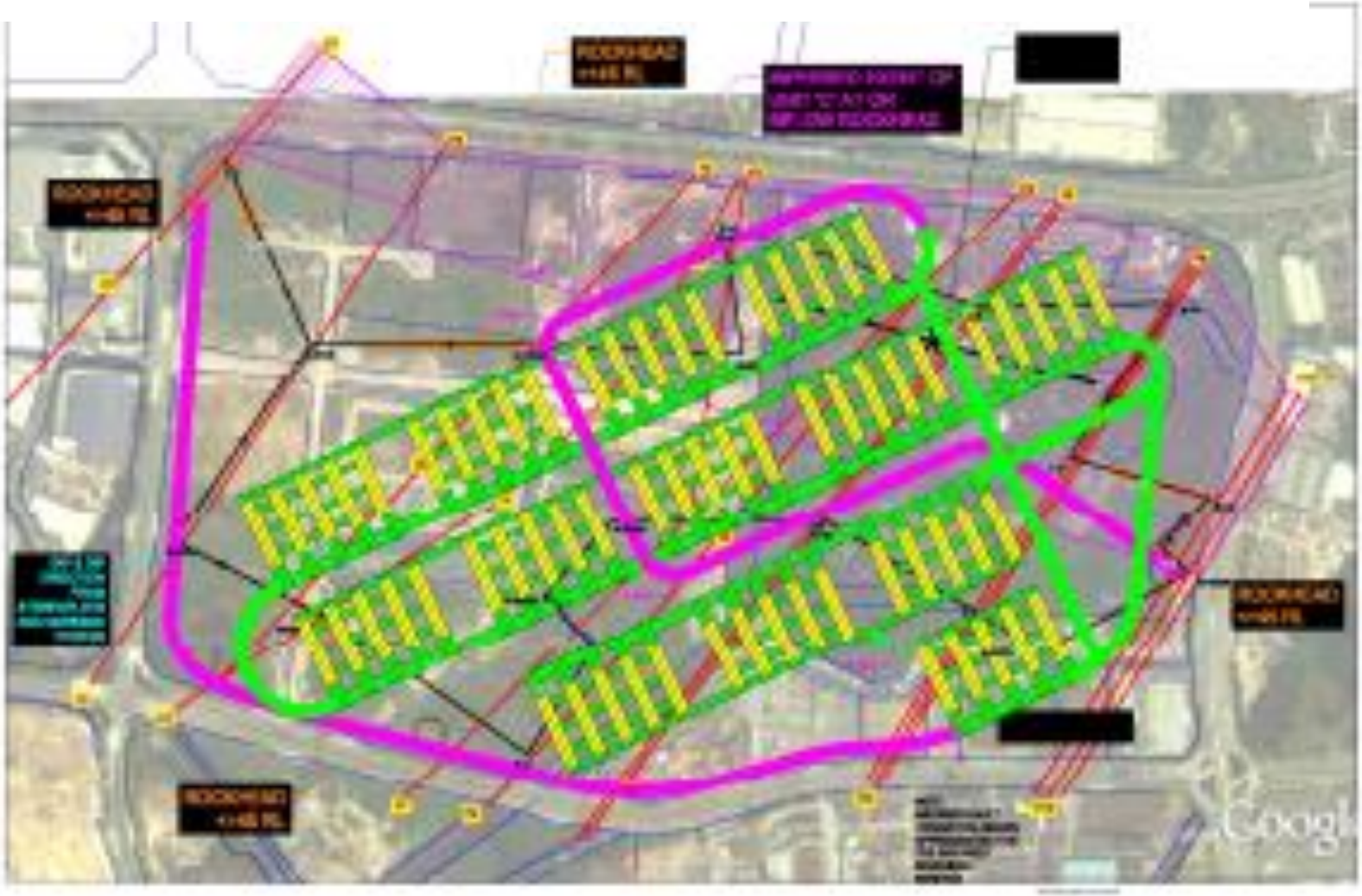
Planned Dedicated Tunnels for Goods



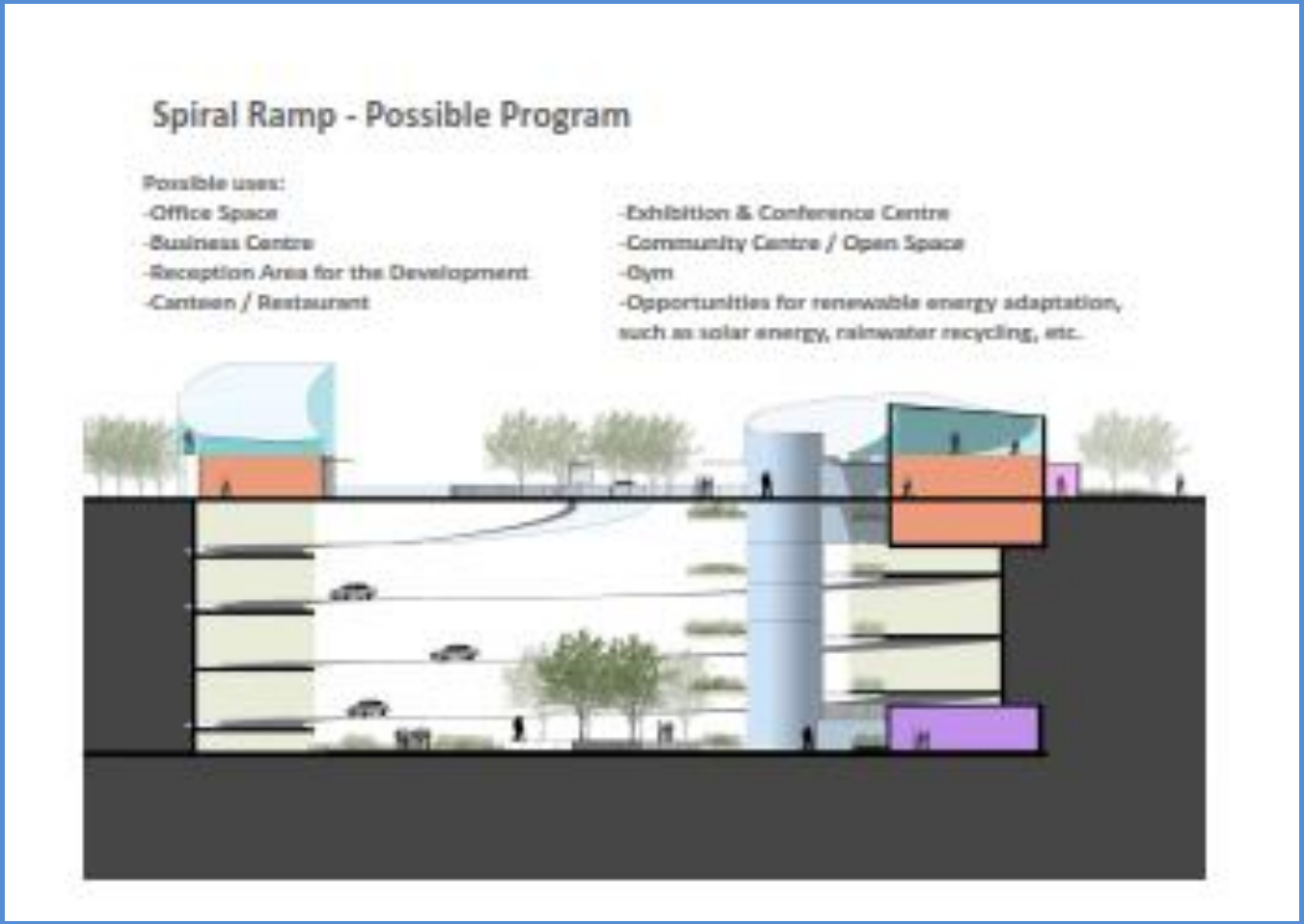
Concept for Capsule Transport of Freight in Pipelines



Preliminary Design : Warehouse and Logisitics Centre



Spiral ramp vehicular access

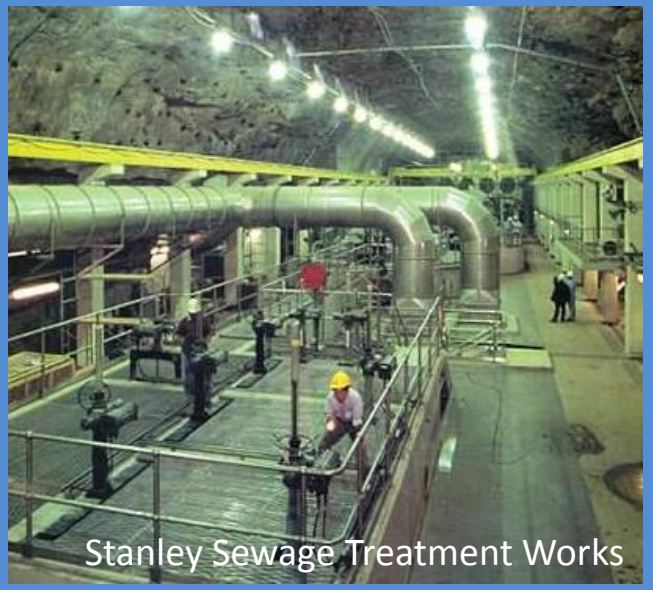


Rock Cavern Developments for Utilities


Island West Refuse Transfer Station


West Island Line Explosives Magazine


HK University Salt Water Reservoir


Stanley Sewage Treatment Works

Planned for Singapore

Sewage Treatment



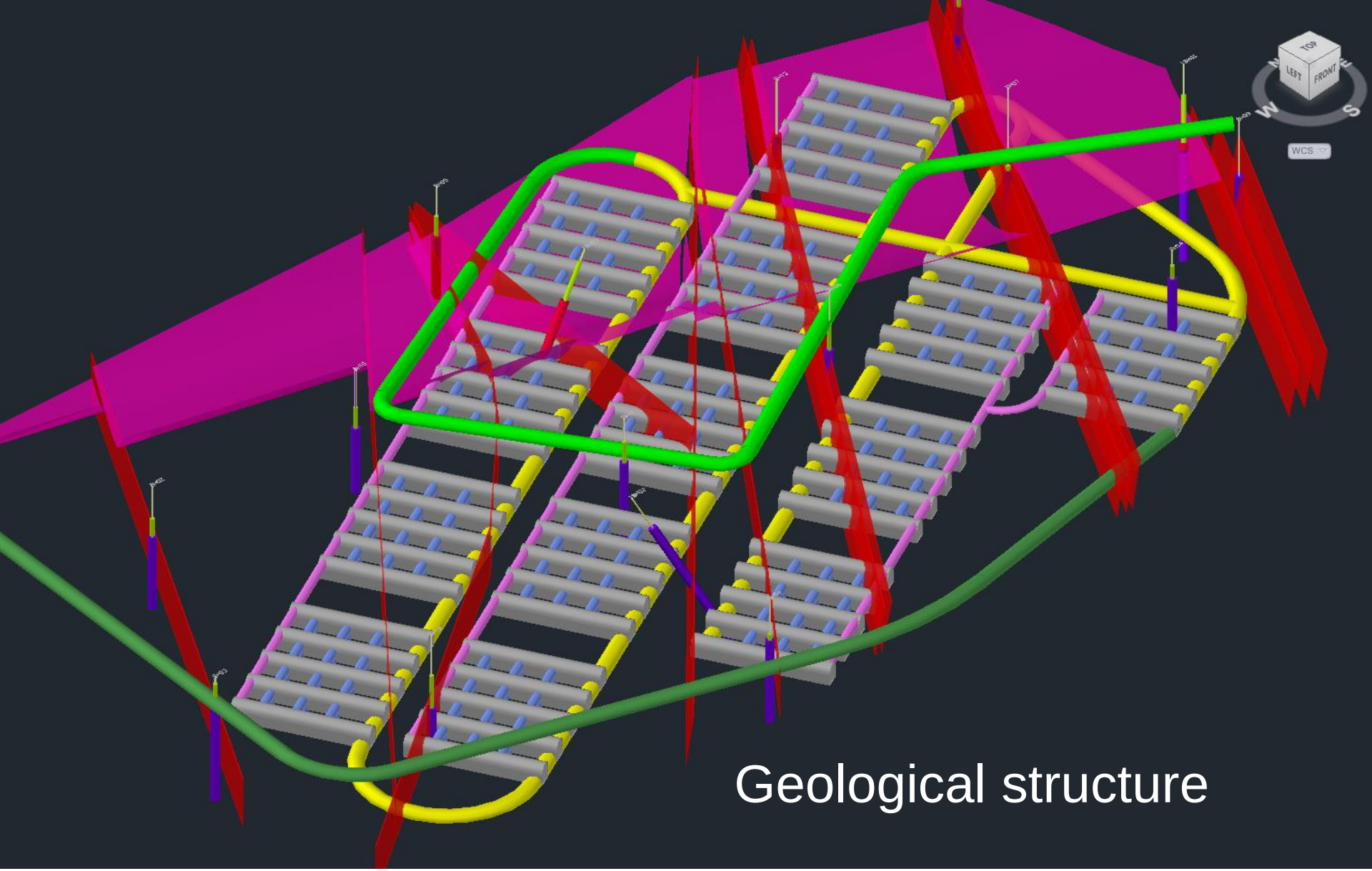
Hong Kong :

Sha Tin Sewage Treatment Works 800,000 people



Put it under the adjacent hill





Geological structure

All that is visible is the access road

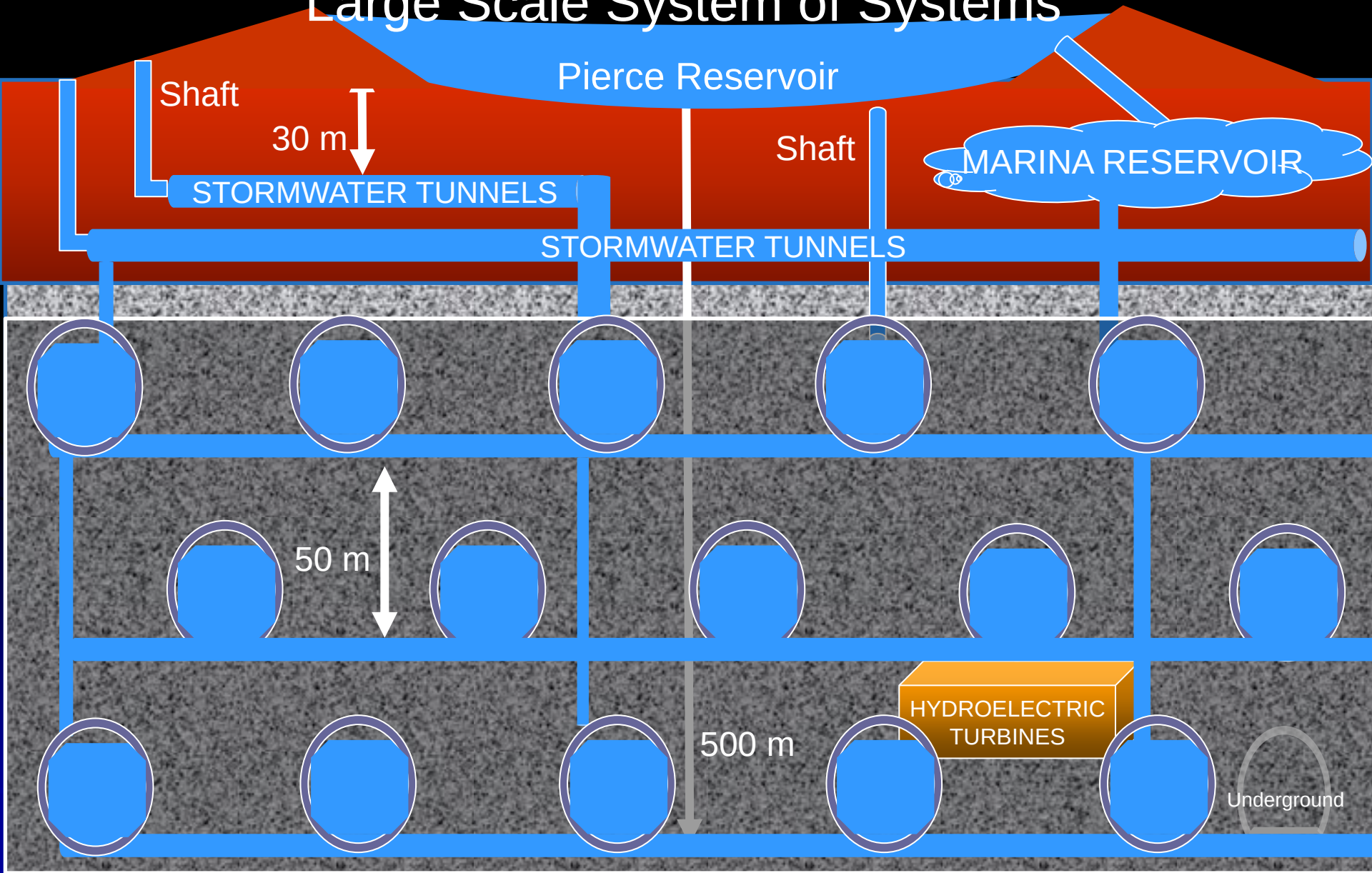


Planned for Singapore

Water storage



Integration of Flood Protection, Storm Water Harvesting and Electricity Storage in a Large Scale System of Systems



Planned : Education below ground

Underground Master Planning for Nanyang Technological University, Singapore [NTU]







NANYANG TECHNOLOGICAL UNIVERSITY

Courtesy of jtc Corporation AECOM

Planned: Education below ground

Underground Science City Kent Ridge Concept



Picture courtesy of JTC

Which ones are below ground?

L6 or B6 button in the elevator









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HONG KONG (CHINA)
CHINA
114°N
114°20'N
22°30'N
22°10'N
Kat O Chau
Luk Keng
Sheung Shui
Plover Cove Country Park
Fanling
Hong Kong Wetland Park
Fairview Park
Plover Cove Reservoir
Tap Mun Chau
Tai Tan Hoi Hap
Tin Shui Wai
Yuen Long
Tai Po
Sai Kung West Country Park
Hung Shui Kiu
Sai Kung East Country Park
Tai Mo Shan Country Park
Tuen Mun
Tsuen Wan
Sha Tin
Sai Kung
Pak Tam Chung
Hong Kong Heritage Museum
Leung Shuen Wan Chau
Tsing Yi
Golden Shopping Arcade
Ladies Market
Kwun Tong
Chek Lap Kok
Hong Kong Disneyland
Kowloon
Hong Kong Science Museum
Kennedy Town
Hong Kong Museum of Art
Tseung Kwan O
Tung Chung
HONG KONG
Madame Tussauds
Hong Kong Museum of Coastal Defence
Hei Ling Chau
Aberdeen
Chai Wan
Lantau
Aberdeen Harbour
Tung Lung Chau
Ocean Park
Fish market, Aberdeen Harbour
Stanley Market
Cheung Chau
Stanley
Shek Kwu Chau
Lamma
Lo Chau
Sai Pok Liu Hoi Hap
Tung Pok Liu Hoi Hap
South China Sea
Po Toi
0
5 Km
0
2 Miles
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Current and Designed Underground Facilities

Strategic Planning Hong Kong

Study of Potential Use of Underground Space 1990

Cavern Project Studies

Preliminary Engineering Geology Studies

Enhanced Used of Underground Space

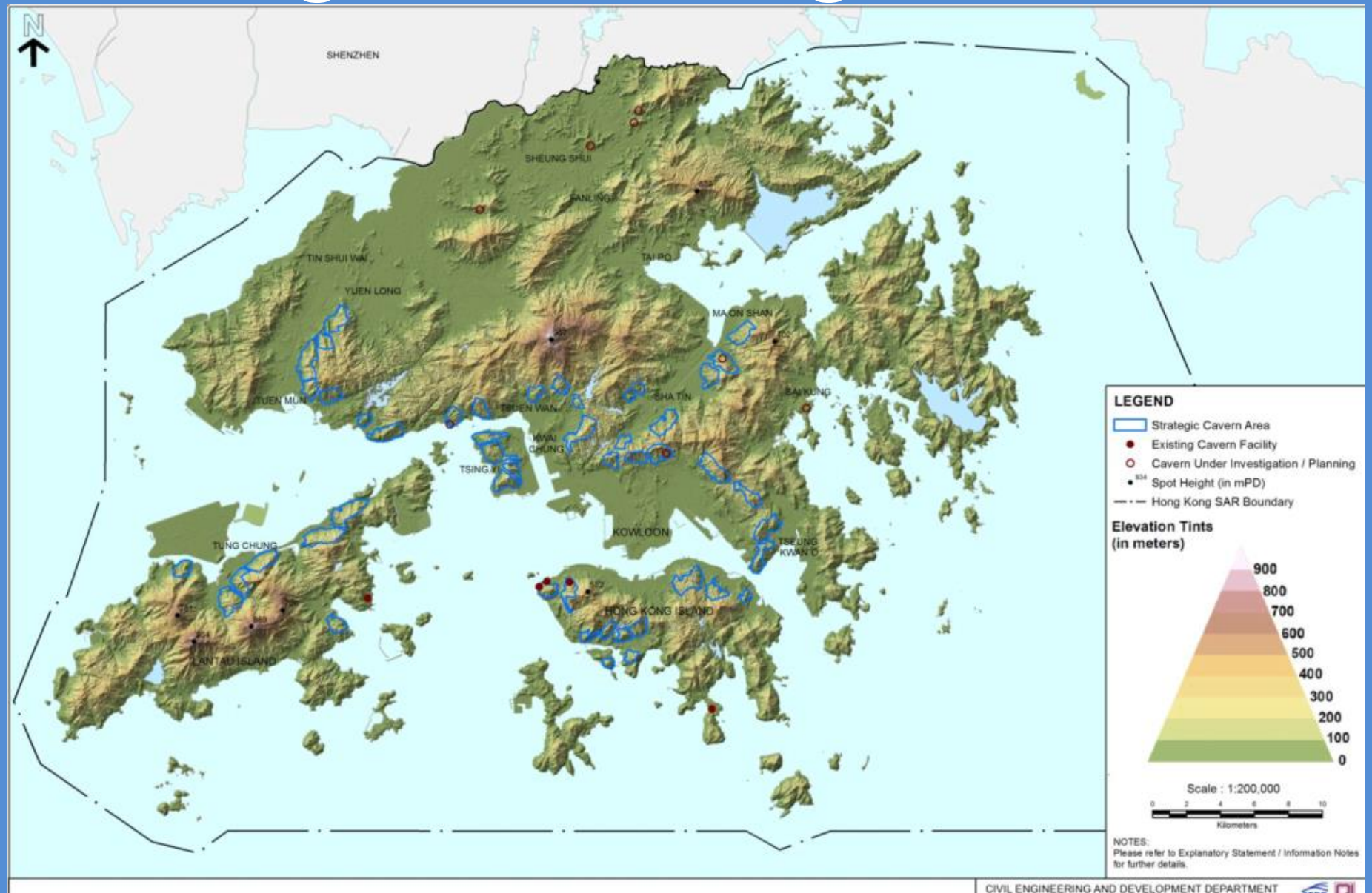
Further Geological Studies

Territory-wide Study on Underground Space

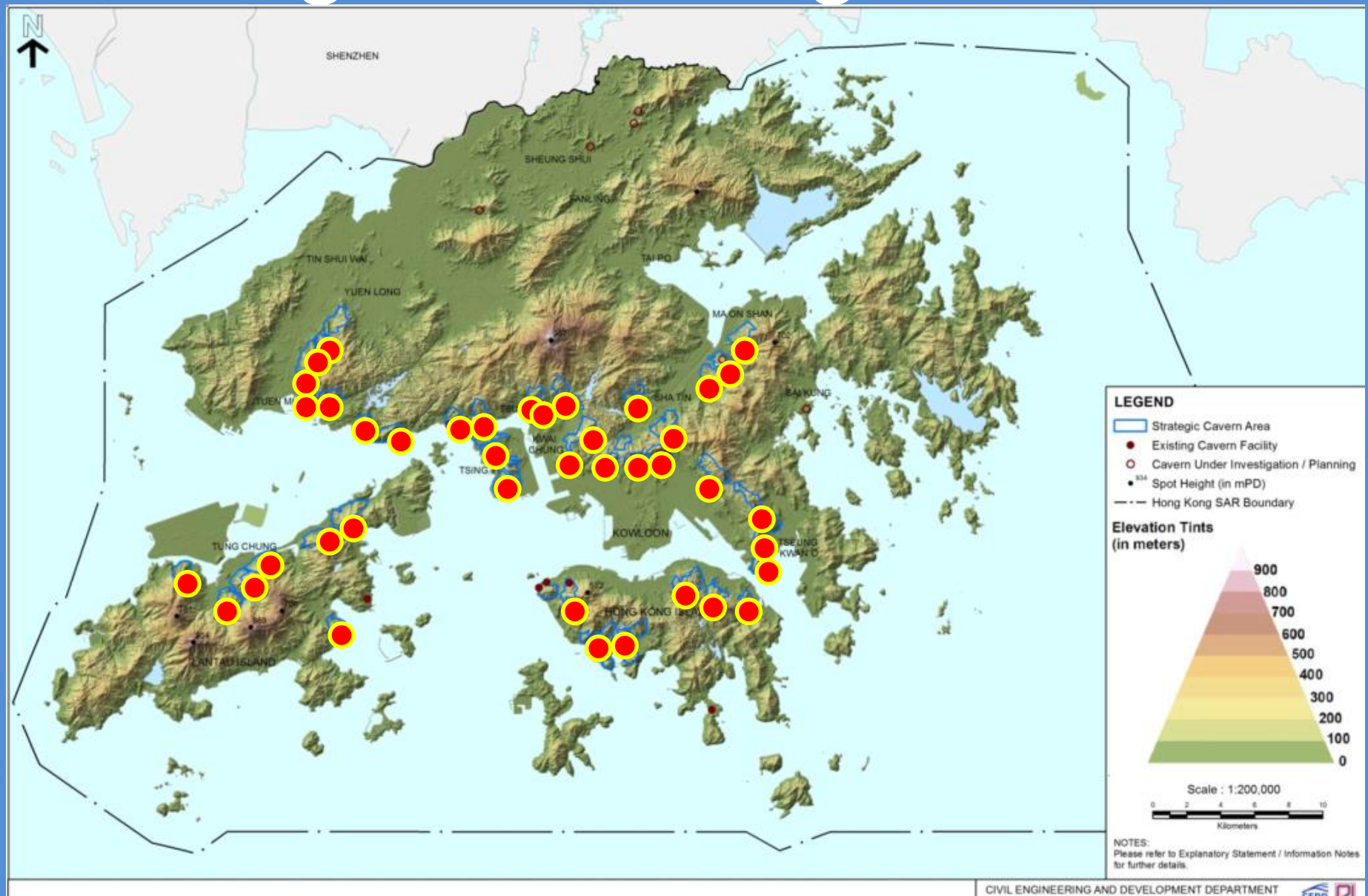
Cavern Master Plan

Pilot Study on Selected Urban Areas 2017

Regional Planning Studies



Regional Planning Studies



Completed Studies

- Technical Study on Underground Quarrying in Hong Kong (Includes both territory wide site search, and also development of a reference scheme based on Lam Tei site)
- Technical Study on Potential Quarry Sites at Tuen Mun West (Includes study of underground quarry as one option)
- Preliminary Land Use Study for Lam Tei Quarry and the Adjoining Area (covers two strategic cavern development areas, and the main underground quarry site being investigated. The study includes consideration of relocating a refuse transfer station and a water treatment works underground).
- Pilot Study on Underground Space Development in Selected Strategic Urban Areas
- Relocation of Sha Tin Sewage Treatment Works to Caverns

New opportunities

Because of Intense uses of surface and large volumes of undeveloped **rock** below

Almost anything can go below ground:-
Water, power transmission, swimming pool, skiing, skating, universities, shopping, restaurants…….

anything that does not need the sky